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Lesson 7: Trees and Ensembles

Technologies for Artificial Intelligence

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20 November 2026

Agenda

- A fourth family, built on a threshold
- Decision trees, and why one alone is mediocre
- Depth as the bias-variance dial, again
- Bagging and random forests
- Feature importance, and its limits
- Gradient boosting
- Choosing among trees, forests and boosting

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Exercise 6, before we start

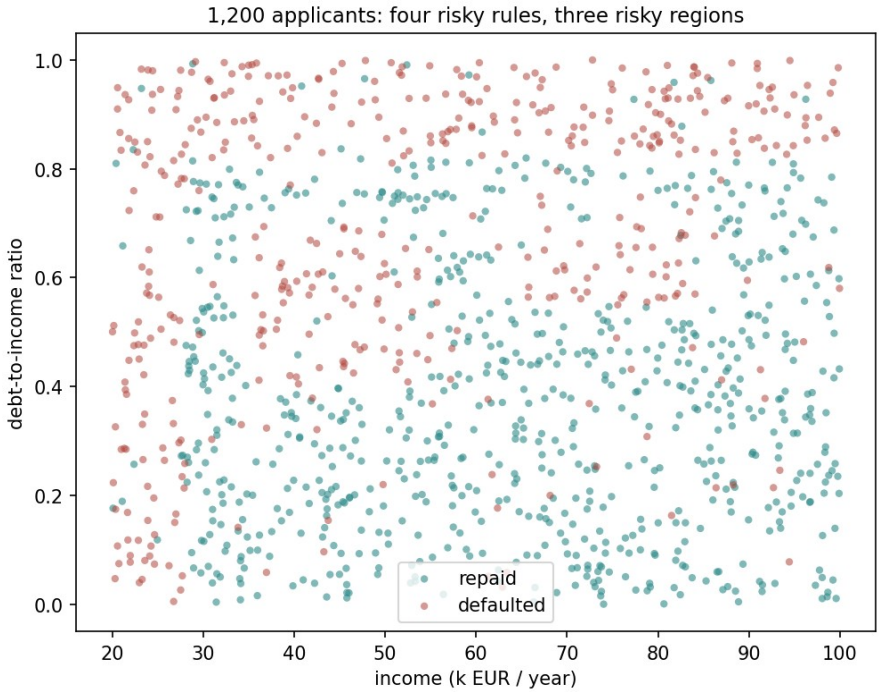
- What counts is the **defence of which family**, not the winning score
- The gap to watch for: a strong accuracy number with no check of *why* the assumption behind it held
- Today’s models come with the same obligation

1,200 loan applicants, two figures each

- Income in thousands of euros a year, and debt-to-income ratio
- Underwriting is a **checklist**: an income floor, a debt ceiling, and two “stressed” combinations, each figure ordinary alone but risky together
- **7%** of labels flipped: no honest model beats about **0.93**, the **noise ceiling**
- **38.7%** defaulted, so the majority baseline is **0.613**

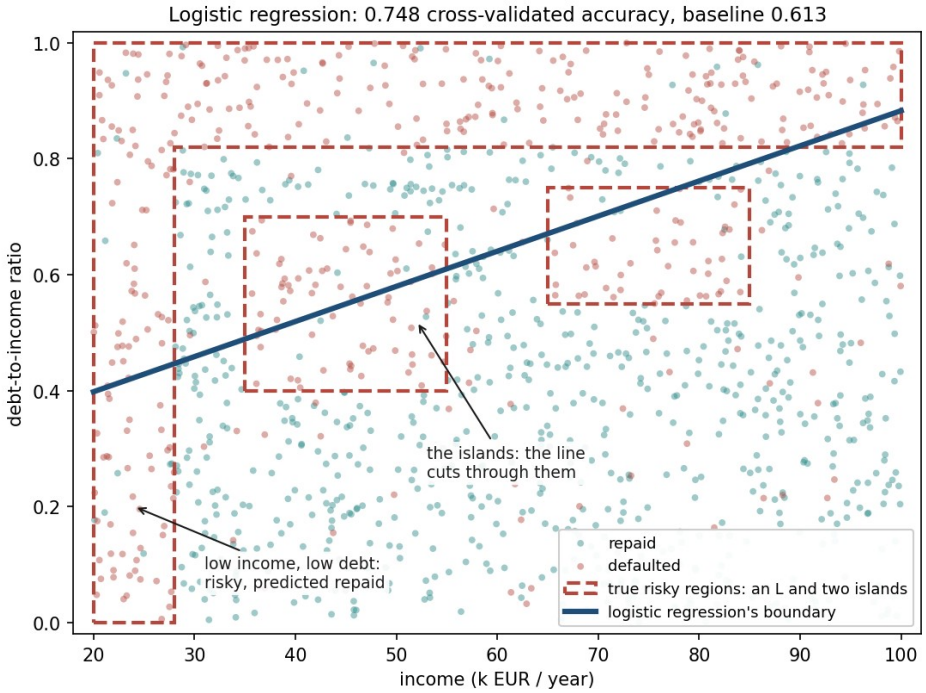
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Four risky rules, three risky regions



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No straight line separates them



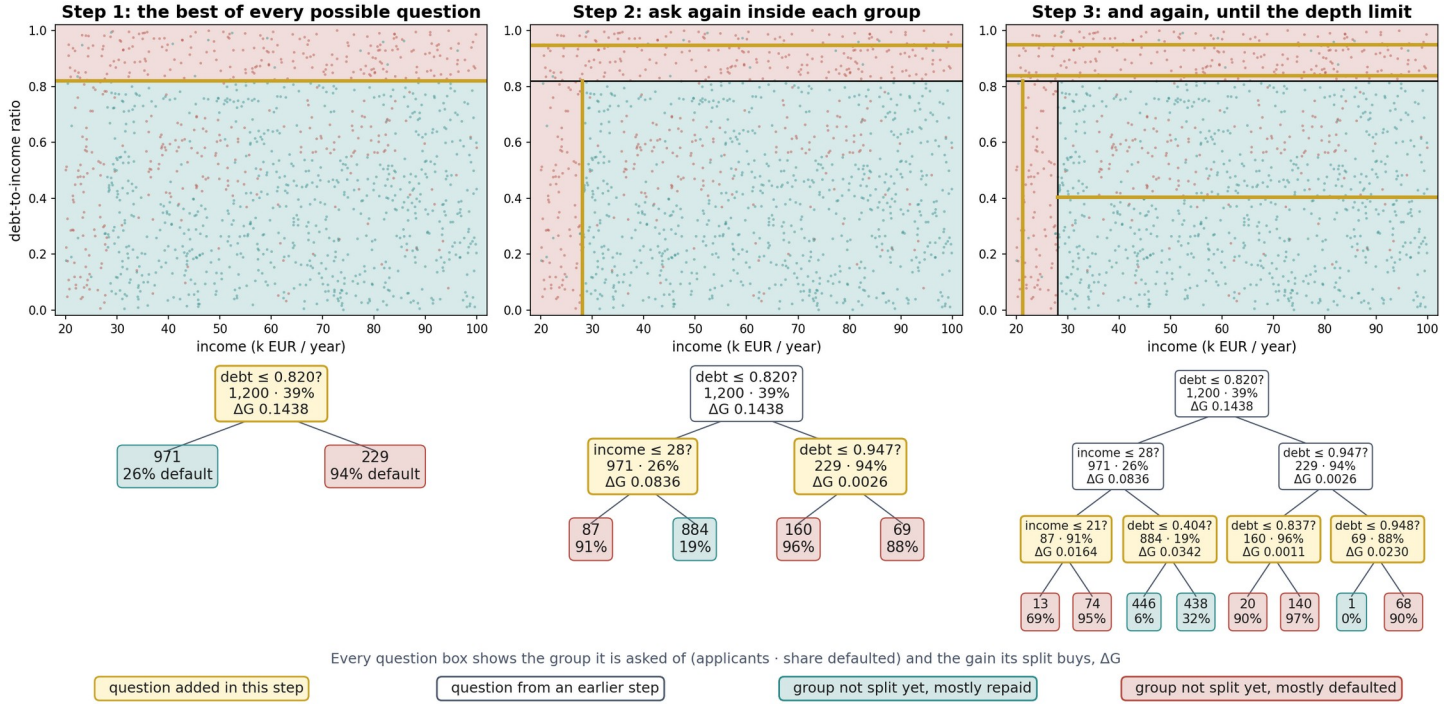
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The whole algorithm, in one sentence

- Try **every feature and every threshold**; keep the split leaving the **purest** two groups
- Apply it; repeat on each group, independently
- Stop at a depth limit, a pure group, or too few examples to split
- No coefficients, no gradient: thresholds are found by **searching**
- **Greedy**: whichever split helps most *right now*, with no lookahead

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The algorithm, drawn



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How pure is a group of examples?

- A leaf that is all one class needs no more questions
- A group half one class, half the other, is the worst case: a coin flip
- **Impurity** is a number that captures this: 0 for a pure group, highest for an even mix. “As pure as possible” then means “with the lowest impurity”

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Gini impurity

$$G = 1 - \sum_{c=1}^{C} p_c^2$$

p_c : share of class c (defaulted, repaid)

slide 8's 229 : $1 - 0.939^2 - 0.061^2 = 0.115$

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For two classes, one number

- G = 2p(1-p), where p is the share defaulted and 1-p the share repaid
- **0** when the group is pure (p = 0 or p = 1)
- **0.5**, the maximum, at p = 0.5
- On the full 1,200 applicants, p = 0.387 gives G = 0.474

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The gain a split buys

- A split divides a parent P into a left child L and a right child R; *m* counts each one's examples: at the root, 1,200 in P = 971 in L + 229 in R
- Worth taking if the children's impurities, **averaged**, are lower than the parent's: at the root 0.474 against 0.330
- The average is **weighted by size**: one example cut off alone would halve a plain average, and barely moves a weighted one

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The reduction in impurity

$$\Delta G = G_P - \left(\frac{m_L}{m_P} G_L + \frac{m_R}{m_P} G_R \right)$$

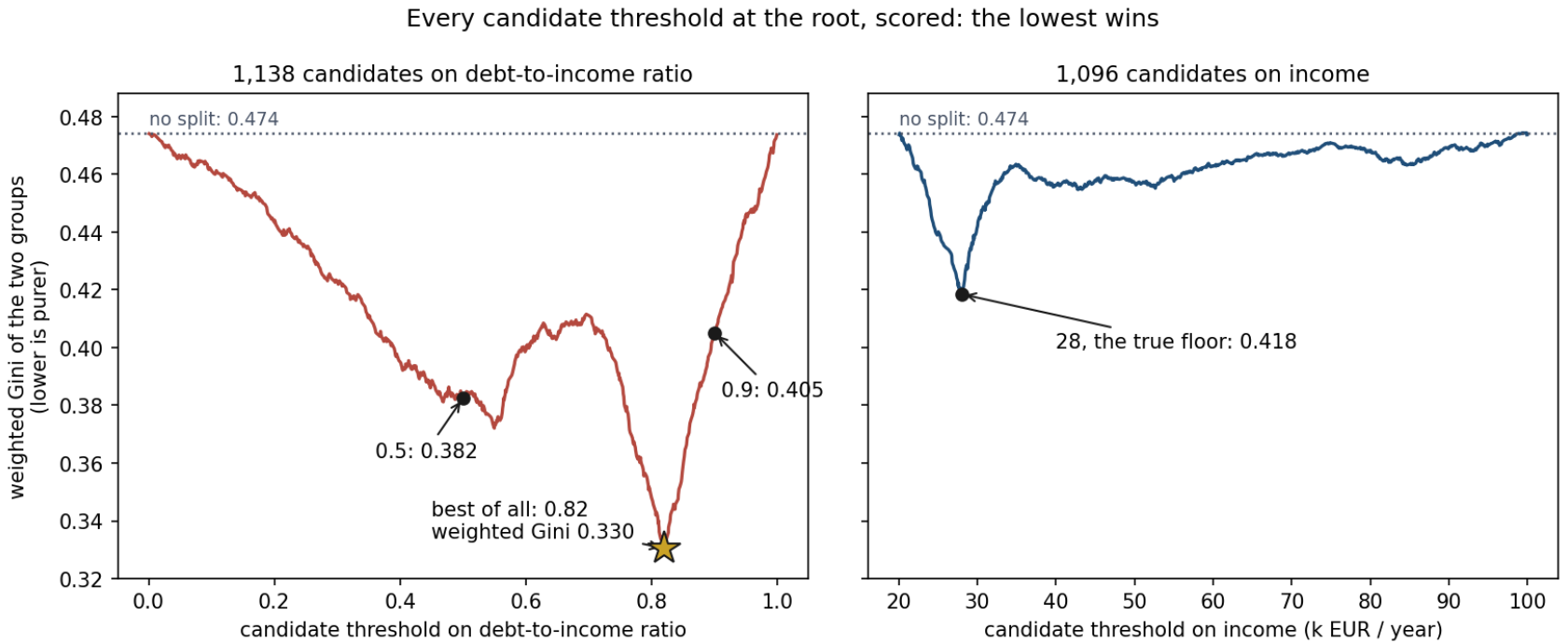
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Scanning for the best split

- Sort a feature’s values: every **midpoint** between neighbours is a candidate threshold
- Infinitely many thresholds, but **finitely many splits**: any threshold between the same two neighbours makes the same two groups
- Score every candidate on **every feature**; keep the best pair, then repeat inside each child
- **Greedy**: ΔG is judged one split at a time, never jointly

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Every threshold, scored



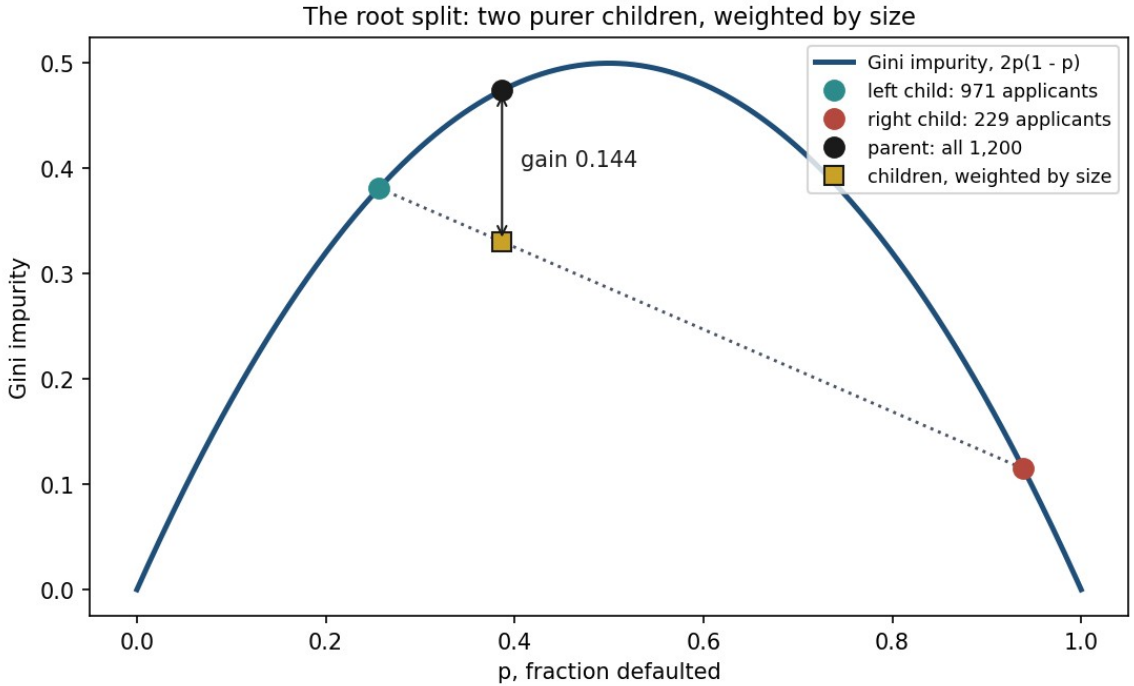
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Worked example: the root split

- Parent: 464 of 1,200 defaulted, G = **0.474**; the search finds `debt_ratio <= 0.82`, the **debt ceiling**
- Left: 249 of 971 defaulted, p = 0.256, G = 2 × 0.256 × 0.744 = **0.381**
- Right: 215 of 229, p = 0.939, G = 2 × 0.939 × 0.061 = **0.115**
- Weighted: 0.809 × 0.381 + 0.191 × 0.115 = 0.330; gain 0.474 - 0.330 = **0.144**

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The root split, on the Gini curve



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The loop, in seven lines

```
def grow(g):
  if stop(g): return leaf(g)
  gain = {c: ΔG(g, c) for c in splits(g)}
  best = max(gain, key=gain.get)
  if gain[best] <= 0: return leaf(g)
  L, R = split(g, best)
  return node(best, grow(L), grow(R))
```

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Checked against scikit-learn

- The from-scratch implementation and scikit-learn's `DecisionTreeClassifier` make **the same prediction for all 1,200 applicants**, at every depth tested
- Not a coincidence: both search the same space by the same greedy rule
- From here on, the notebooks use scikit-learn's implementation: the same algorithm, compiled, not a different one

What greedy misses

- The search never asks “which pair of splits helps most together”: only “which single split helps most right now”
- A feature that only pays off **in combination** with a later split can be invisible to a shallow tree
- Exactly what happens to this dataset’s two interior “stressed” islands, covered next

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Depth is the bias-variance dial

- `max_depth`: the most questions **one applicant** answers
- **Shallow tree**: few questions, only the coarsest structure. High bias, low variance
- **Deep tree**: enough questions to isolate single points. Training accuracy **exactly 1.000**, even on data with no structure, unless two examples share every feature value but not the label

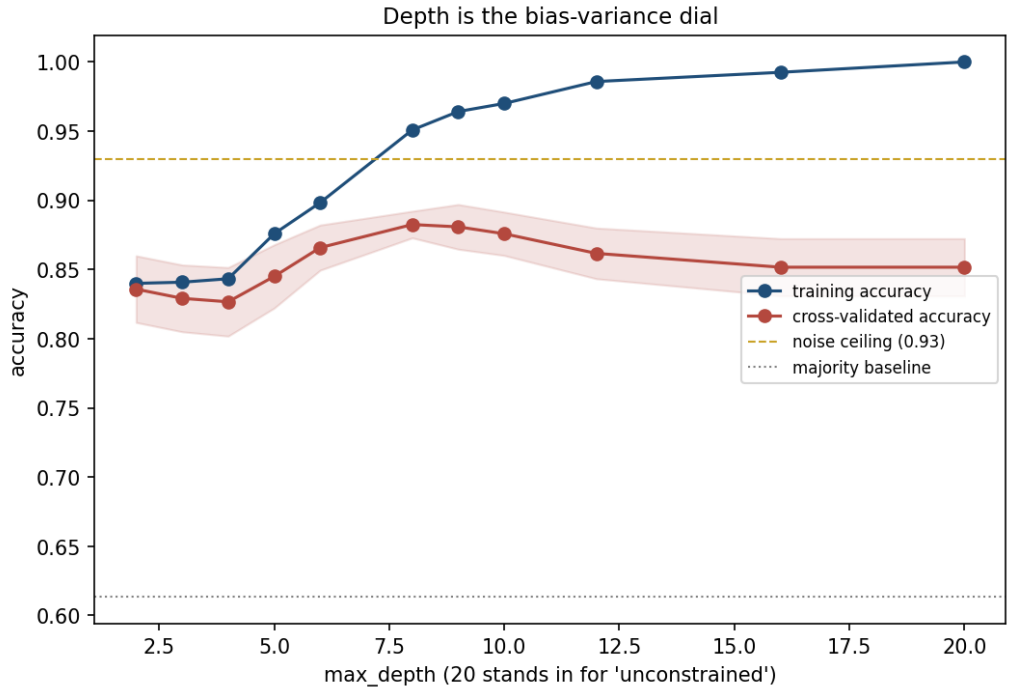
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Measured, at seven depths

`max_depth`	Training accuracy	Cross-validated, 5 folds (mean ± standard deviation)
2	0.840	0.836 ± 0.024
4	0.843	0.827 ± 0.025
6	0.898	0.866 ± 0.016
8	0.951	**0.882 ± 0.010**
12	0.986	0.862 ± 0.018
16	0.993	0.852 ± 0.021
unconstrained	**1.000**	0.852 ± 0.021

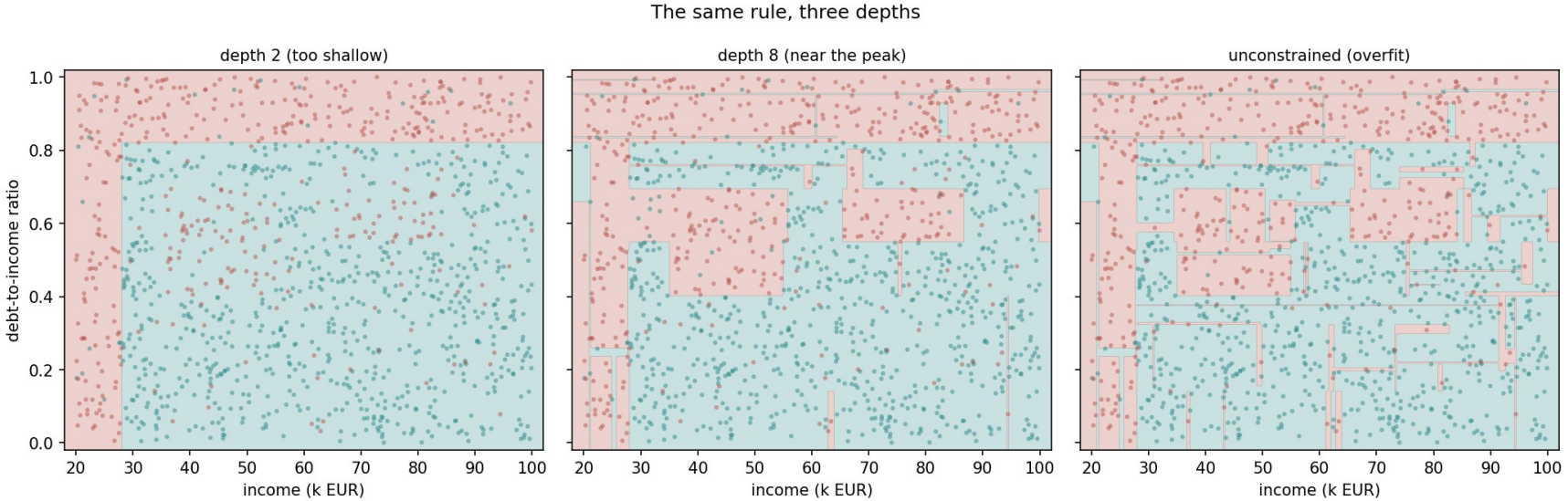
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The curve bends downward



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The same rule, three depths



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Reading a tree, and the knob that keeps it small

- `max_depth` **caps every path**: each node above the cap is split if it can be, worth it or not
- `max_leaf_nodes` is a **budget of leaves**, spent on the largest ΔG first
- `max_leaf_nodes = 9`: **0.908** training, **0.901** cross-validated, above depth 8's 0.882; thresholds 28 and 0.82
- In code: `export_text(tree)` prints the tree as if/else rules

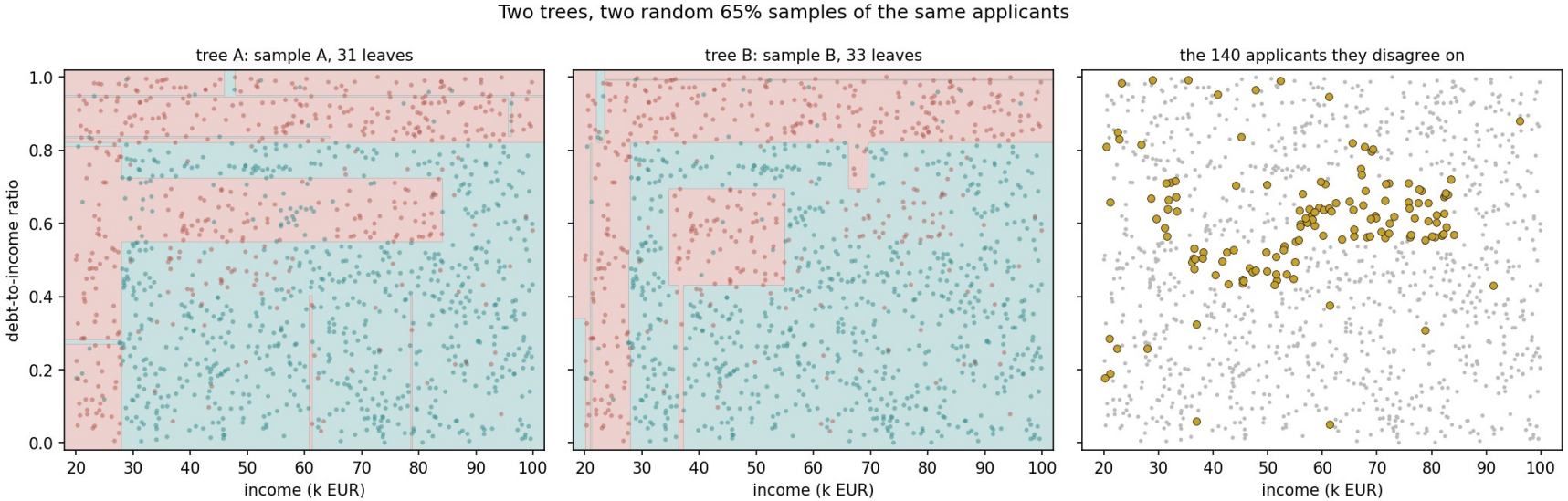
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A tree that doesn’t stay still

- Two trees at `max_depth = 6`, each on 780 of the 1,200 applicants, drawn **without replacement** (no one twice)
- The debt ceiling, the strongest split, agrees to within **0.001**
- They grow **31 and 33 leaves**, and give **opposite verdicts for 140** of the 1,200 applicants (11.7%)
- The strongest split is stable; the weaker ones, which shape most of the tree, are not

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Two trees, 140 different verdicts



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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
1. look at it first, 4-6	`make_pipeline(StandardScaler(), LogisticRegression())` in `cross_val_score(..., cv=StratifiedKFold(5))` → 0.613 against 0.748
2. a tree, split by split, 7-20	your own `gini` and `best_split`, then `DecisionTreeClassifier(max_depth=d)` → 100.0% identical predictions; gain 0.1438
3. the depth dial, 21-24	`DecisionTreeClassifier(max_depth=d)` in `cross_val_score` → peak 0.882 at depth 8
4. what a tree costs, 25-27	`DecisionTreeClassifier(max_leaf_nodes=9)`, `export_text` → 0.908 training, 0.901 cross-validated; two trees, 140 opposite verdicts

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Break

- Twelve minutes

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Bagging: averaging away the instability

- Trees differ across samples, so **many trees, averaged** should partly cancel their mistakes
- **Bootstrap sample**: m rows drawn **with replacement** from the m training rows
- One tree per sample, `n_estimators` times; predict by **majority vote**. That is **bagging**: bootstrap aggregating
- In code: `BaggingClassifier()`, whose default model is a decision tree

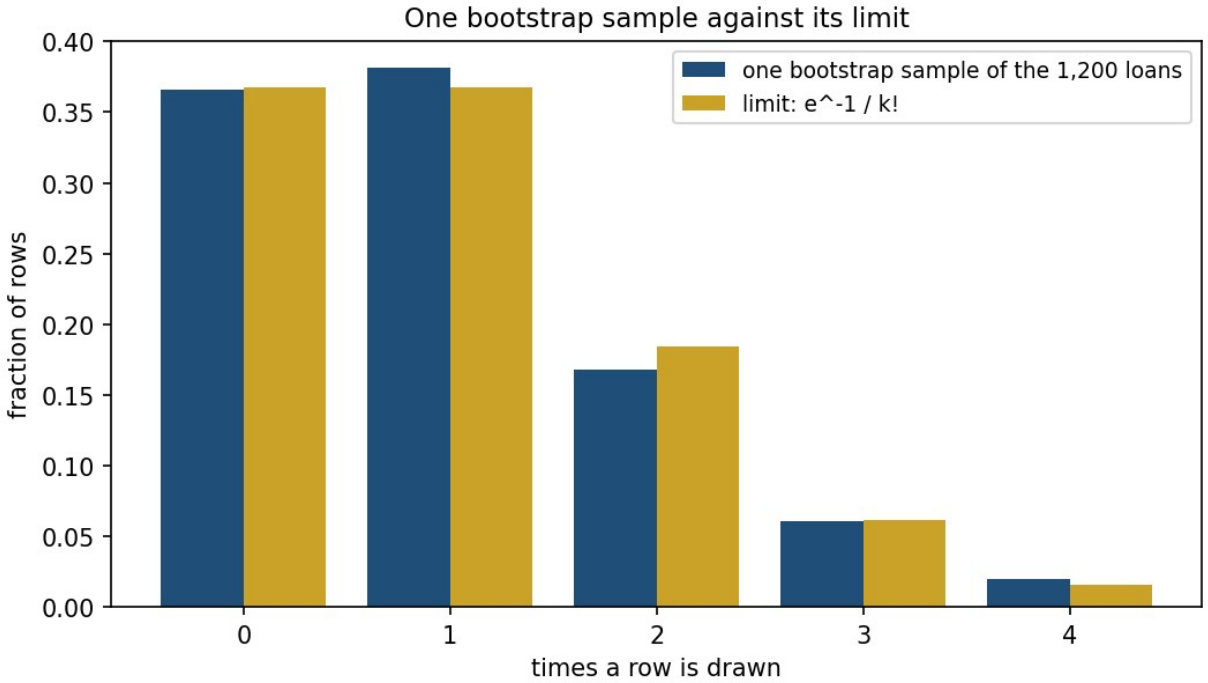
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How much data one bootstrap sample sees

- One row is missed by all m draws with probability $(1 - 1/m)^m$, which tends to $e^{-1} \approx 0.368$ as m grows
- So about **63.2%** of the distinct rows are drawn at least once
- The remaining **36.8%** are left out entirely: called **out-of-bag (OOB)**
- At m = 1,200: a single simulated draw left out 36.6% of rows, close to the limit

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One bootstrap sample, counted



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Out-of-bag: free validation

- Predict each row by a vote of only the trees that never saw it: the accuracy of those votes is the **OOB score**, with no rows set aside
- A 300-tree random forest (bagged trees plus one rule, slide 37): OOB **0.9117**; 5-fold cross-validation **0.9117 ± 0.021**
- The four-decimal match is **luck**: over twelve seeds the two differ by 0.0026 on average, well inside ±0.021

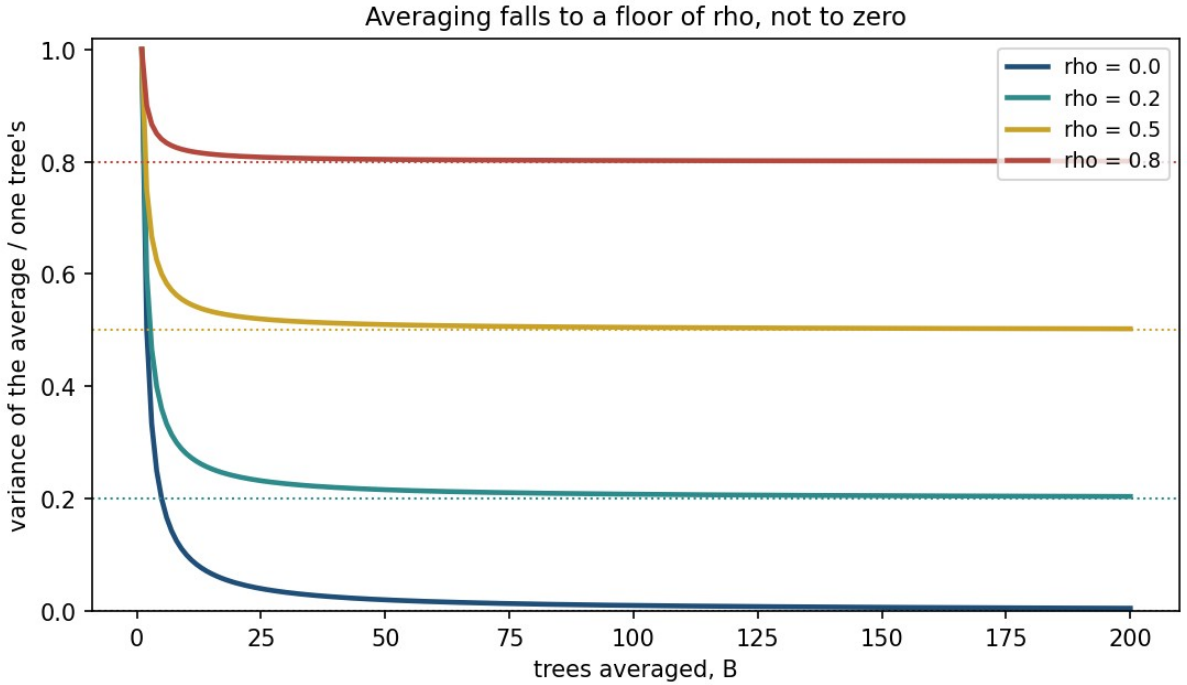
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Why averaging reduces variance

- Model each tree’s prediction as noisy, with variance σ^2 over the bootstrap sample it drew, and let ρ be the correlation between two trees
- The average of B trees has variance $\sigma^2/B + \rho\sigma^2 (B-1)/B$: a **floor** of $\rho\sigma^2$, **not zero**: more trees soon buy little
- Averaging **cannot repair an error every tree shares**: trees too shallow for the rule stay too shallow, however many are averaged

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Averaging stops at a floor



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Bagging, measured

Model	Mean test accuracy, 30 random 70/30 splits	Standard deviation across the 30
Single unconstrained tree	0.856	0.0185
Bagging, 100 trees	**0.902**	**0.0153**

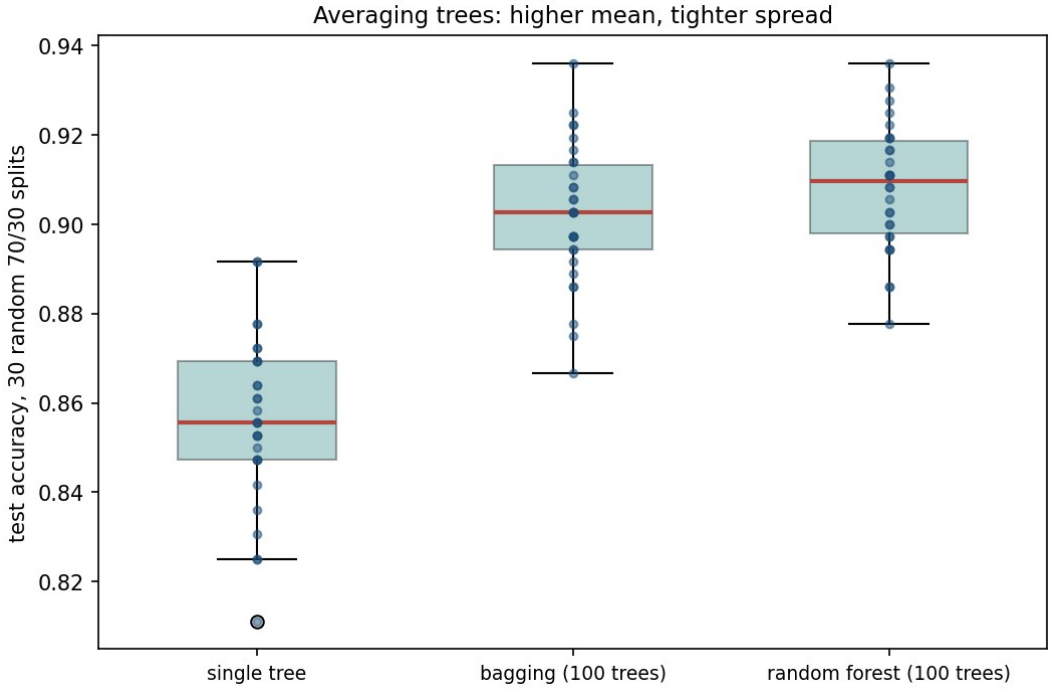
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Random forests: decorrelating the trees

- The floor is $\rho\sigma^2$: lowering it means making the trees agree **less**
- Bagged trees see **every feature**, so a strong one pulls them all the same way
- A **random forest** offers each split a random subset of `max_features` (default $\sqrt{n}$, rounded down), lowering ρ
- In code: `RandomForestClassifier()`, 100 trees by default

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Random forests, measured



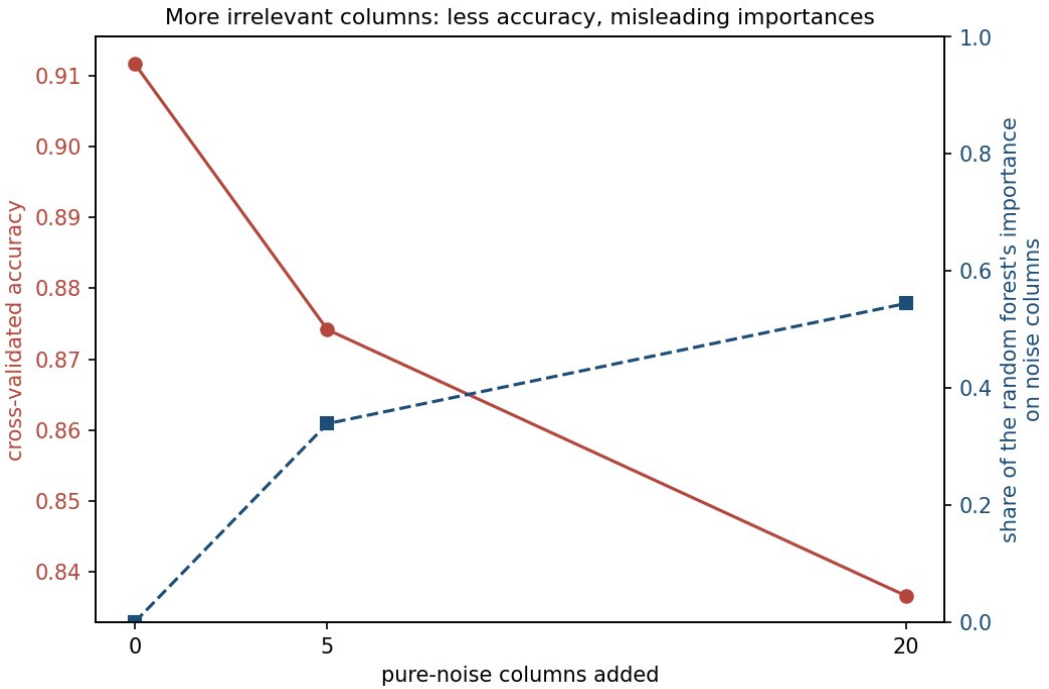
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Feature importance: what the forest claims to tell you

- **Feature importance**: the impurity reduction credited to each feature over every tree and split, scaled to add up to 1
- With only the two real features here, it names exactly those two
- What happens once we add columns of random numbers, unrelated to default?
- In code: `forest.feature_importances_`

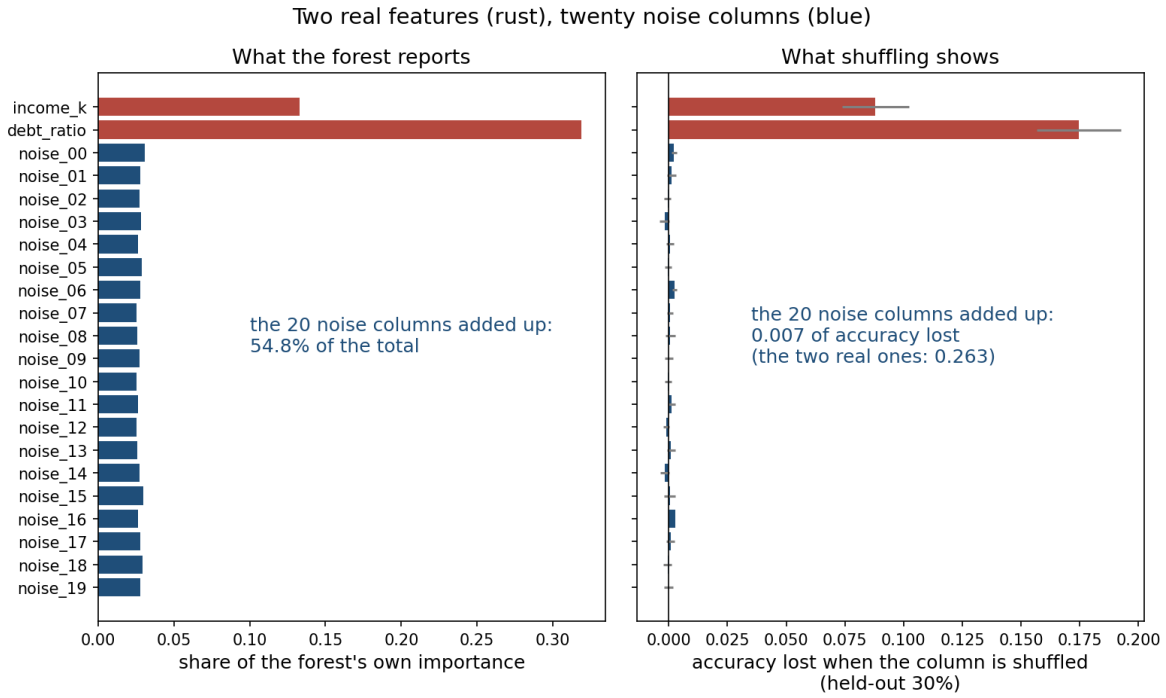
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More columns, more noise



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Permutation importance



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Notebook 2, live

section, and its slides	you call (→ the number to come back with)
1. bagging from scratch, 30	`train_test_split`, your bagging against `BaggingClassifier`, `accuracy_score` → 0.872 and 0.872
2. why averaging works, 31-35	`oob_score=True`: `.oob_score_` against `cross_val_score` → 0.9117 both; 0.0026 over 12 seeds
3. variance, measured, 36-38	tree, `BaggingClassifier`, `RandomForestClassifier`, 30 splits → spread 0.0185, 0.0153, 0.0137
5. feature importance, 39-40	`.feature_importances_`, 0 to 20 noise columns → 54.5% on noise
6. checking the importances, 41	`permutation_importance`, test split → real 0.09 and 0.18, noise ≈ 0

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Gradient boosting: correcting mistakes in sequence

- Bagging and random forests build trees **independently**, then average
- **Boosting** builds them **in sequence**: each new tree corrects what the trees so far got wrong
- Each is a shallow **regression tree**, a **weak learner**: it predicts a number, a correction, not a class
- The learning rate α sets **how much** of each correction is applied

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Building the ensemble, one tree at a time, for squared error

$$F_0(x) = \bar{y}, \qquad F_t(x) = F_{t-1}(x) + \alpha \cdot h_t(x)$$

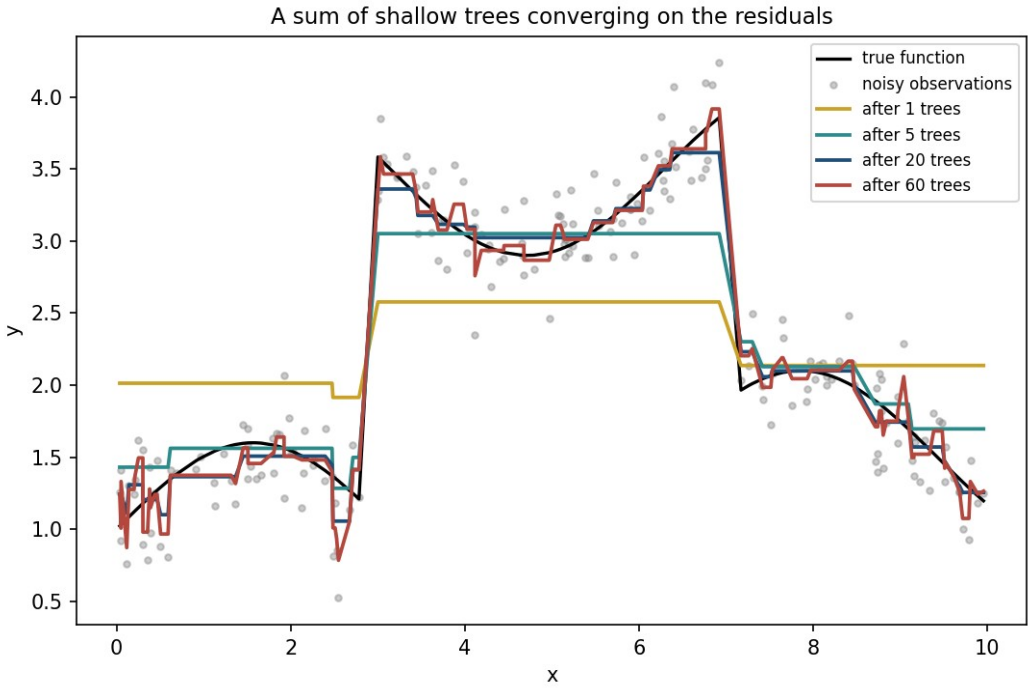
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What each tree is fit to

- For squared error, what remains unexplained is the **residual**, $y - F_{t-1}(x)$
- In general, **functional gradient descent**: h_t approximates the negative gradient of the loss with respect to F_{t-1}
- With $L = \frac{1}{2}(y - F)^2$, $-\partial L/\partial F = y - F$: the two coincide
- In code: `DecisionTreeRegressor(max_depth=2)`, fit to `y - prediction`

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A step plus a sine wave, fit by 60 shallow trees



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Reading the picture

- **1 tree**: mean squared error (MSE) against the true function 0.395, barely moves the flat guess
- **5 trees**: MSE 0.068, the coarse shape appears
- **20 trees**: MSE **0.012**, the ensemble’s best point
- **60 trees**: MSE rises back to 0.018, tracking noise, not signal

Classification: the same idea, a different gradient

- y is 0 or 1; the score F becomes a probability through lesson 4’s logistic function, $p = \sigma(F)$ (this σ is not the σ^2 of the variance slides)
- The loss is lesson 4’s **log-loss**
- Each tree fits the **pseudo-residual** y - p, the negative gradient: lesson 4’s error term
- Start: $p = 0.387$ for all, so the first tree fits $+0.613$ to defaulters and -0.387 to the rest

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On the loan data

- `GradientBoostingClassifier`, defaults: **100 trees**, learning rate $\alpha = 0.1$
- Reaches **0.902 ± 0.020** cross-validated accuracy
- Already ahead of any single tree from earlier in the lesson

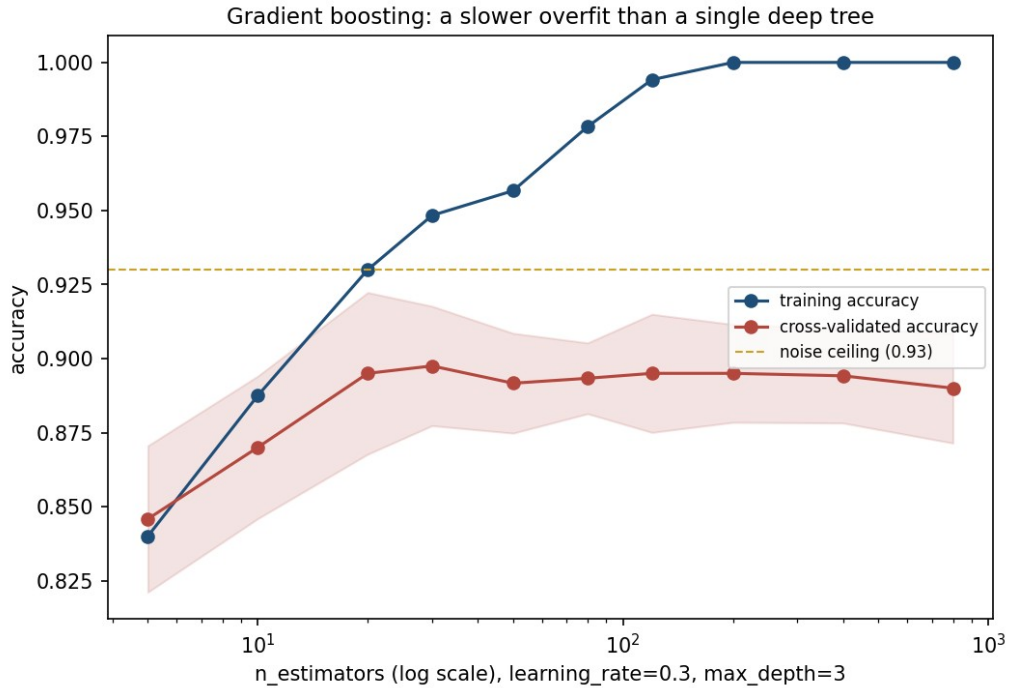
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Boosting never stops fitting on its own

`n_estimators` (α = 0.3, depth 3)	Training accuracy	Cross-validated accuracy
5	0.840	0.846 ± 0.025
10	0.888	0.870 ± 0.024
20	0.930	0.895 ± 0.027
30	0.948	**0.897 ± 0.020**
80	0.978	0.893 ± 0.012
200	1.000	0.895 ± 0.017
800	1.000	0.890 ± 0.019

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Training climbs to 1.000 and stays; the honest curve turns over first



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Fixing 120 trees, varying only the learning rate

Learning rate α	Training accuracy	Cross-validated accuracy
0.02	0.888	0.872 ± 0.025
0.05	0.914	0.897 ± 0.022
0.10	0.939	**0.902 ± 0.022**
0.30	0.994	0.895 ± 0.020
1.00	1.000	0.878 ± 0.020

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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
1. boosting from scratch, 43-47	`DecisionTreeRegressor(max_depth=2)` on the residuals, 60 times → MSE 0.0119 at 20 trees
2. classification, 48-49	`GradientBoostingClassifier()` in `cross_val_score` → F0 = −0.461; 0.902
3. more trees, 50-51	the same, `learning_rate=0.3`, `n_estimators` from 5 to 800 → peak 0.897 at 30
4. learning rate, 52	the same with `n_estimators=120` and `learning_rate` from 0.02 to 1.0 → best 0.902 at 0.1
5. leaderboard, 54	every model in one `cross_val_score` loop → random forest 0.911

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Every method, on the same five folds

Model	Cross-validated accuracy
Majority baseline	0.613
Logistic regression	0.748
Single tree, unconstrained	0.852
Single tree, `max_depth = 8`	0.882
Gradient boosting, 30 trees	0.897
Single tree, `max_leaf_nodes = 9`	0.901
Bagging, 100 trees	0.904
Random forest, 100 trees	**0.911**
noise ceiling	*≈ 0.93*

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Three findings

- **On this dataset**, the ensembles land within 1.5 points of each other: **whether you ensemble** matters more than **which**
- A **nine-leaf tree** reaches 0.901, a point from bagging: on rectangles a small tree is enough, and a person can read it
- The **unconstrained tree is the worst**, though the most flexible

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How to choose, in practice

Situation	Reach for
The decision must be explained to the person it affects	A single tree, small enough to read; if even that is too opaque, none of these models
Tabular data, accuracy matters most, no explanation required	A random forest: the best score here, at its default settings
Accuracy matters, and there is time to tune the number of trees and the learning rate together	Gradient boosting, tuned on validation data
Many columns, few of them informative	Any tree method, but check its importances by permutation
Too few rows to set a validation set aside	A random forest, judged by its out-of-bag score

54% of the importance landed on noise

- 20 noise columns, 2 real ones: not small, not a bug
- **The menu**: 4 of 22 columns per split; in two splits of three (0.662), no real feature
- **Small groups**: with all 22 on offer, noise still takes **28%**, from deep splits
- The check: **permutation importance**, slide 41

What to take away

- A tree splits greedily on Gini impurity, matching scikit-learn
- Depth is the bias-variance dial: accuracy peaked at **depth 8**
- Bootstrap resampling leaves **36.8%** of rows out-of-bag: free validation
- **54% of a random forest’s importance landed on 20 noise columns**, today’s number

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Homework

- **Exercise 7**, discussed at the start of **Friday 27 November**
- `Exercises/07_trees_and_ensembles.md`

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